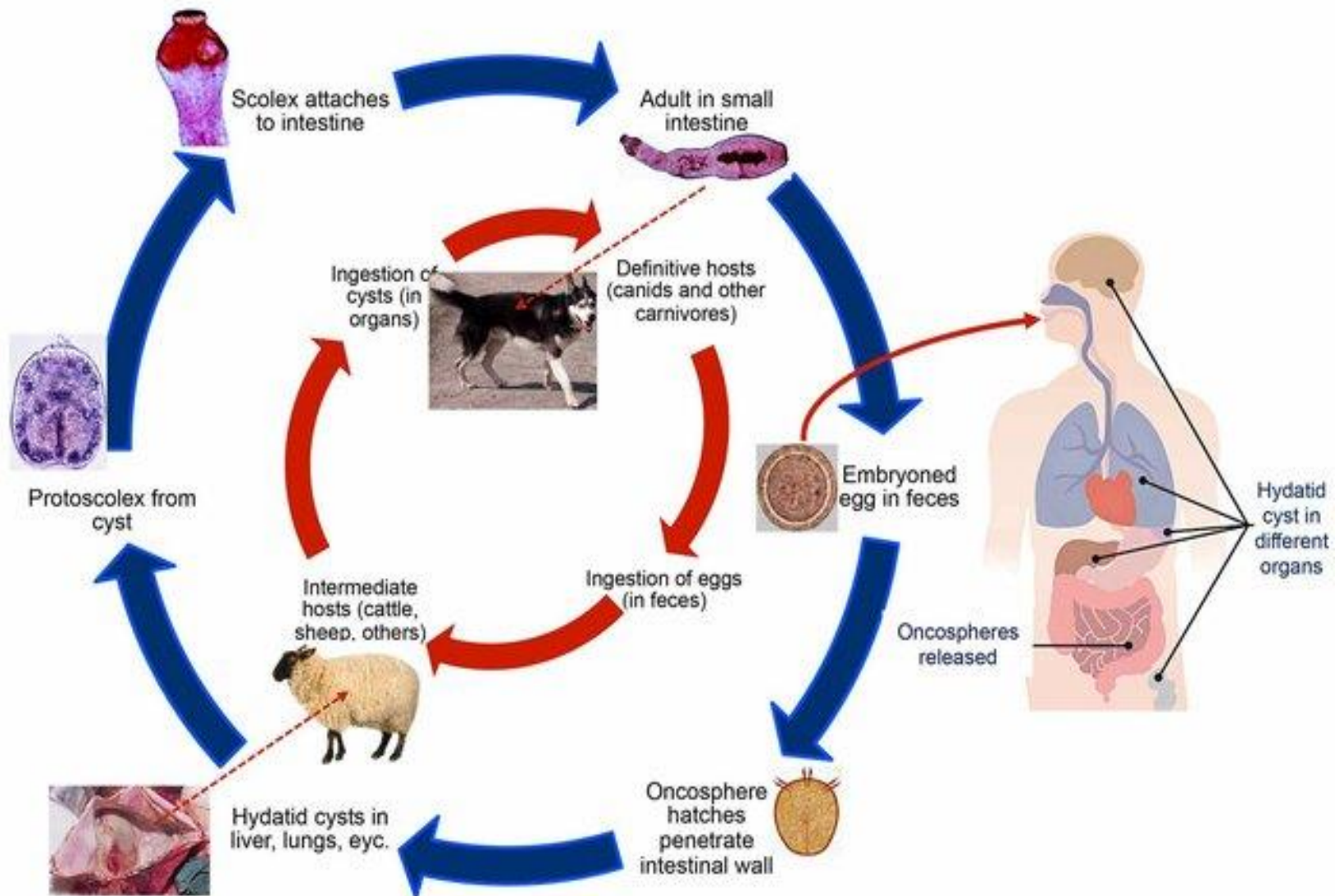


Hydatid disease

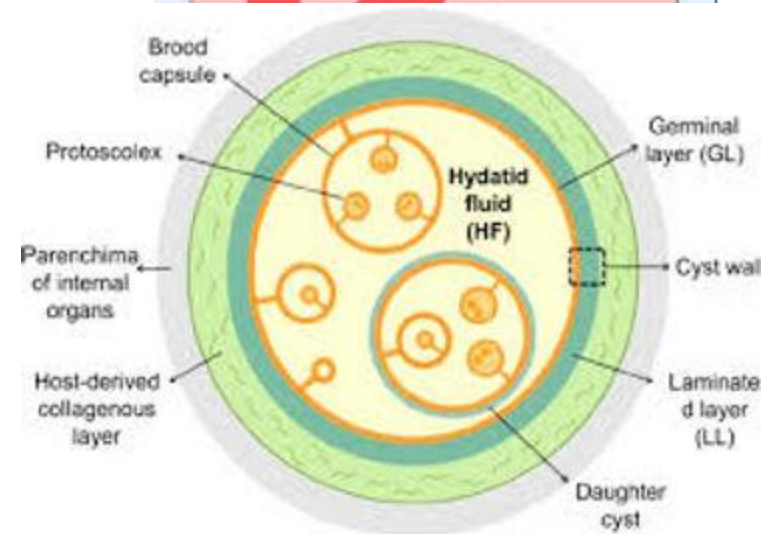
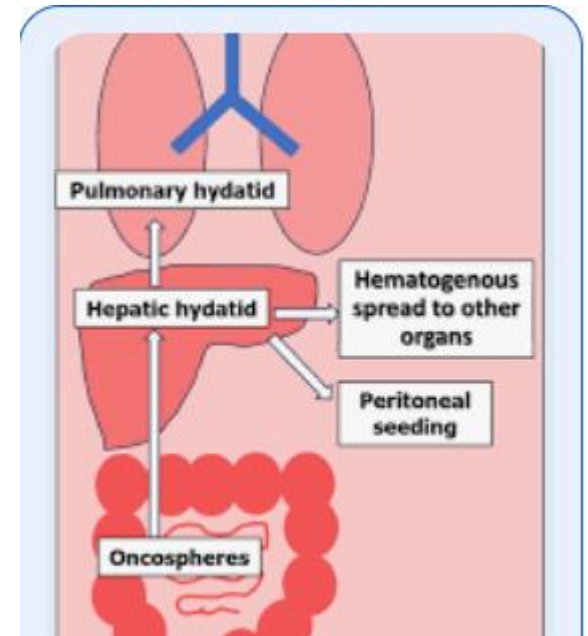
- Hydatidosis results from a larval infection of the tapeworm (cestode) *Echinococcus granulosus*. The disease characteristically demonstrates the growth of hydatid cysts (metacestode) in internal organs of intermediate hosts, including humans.
- The definitive hosts of the cestode are carnivores such as dogs. Humans and other hosts ingest eggs or gravid proglottids that are excreted in the definitive host's faeces, causing the infection
- **Transmission**
- As humans ingest the eggs of the tapeworm, an oncosphere larva is released from the egg, which can penetrate the lamina propria of the intestine





Pathophysiology

- Once it penetrates, it gets passively transported via blood or lymphatics to the liver, lungs, or other internal organs, where they develop into a hydatid cyst (metacestode larvae).
- In humans, cysts grow slowly and can be up to multiple liters in volume and contain thousands of protoscolices.
- Over time, septations and daughter cysts disrupt the typical unilocular pattern of echinococcal cysts.



Histopathology

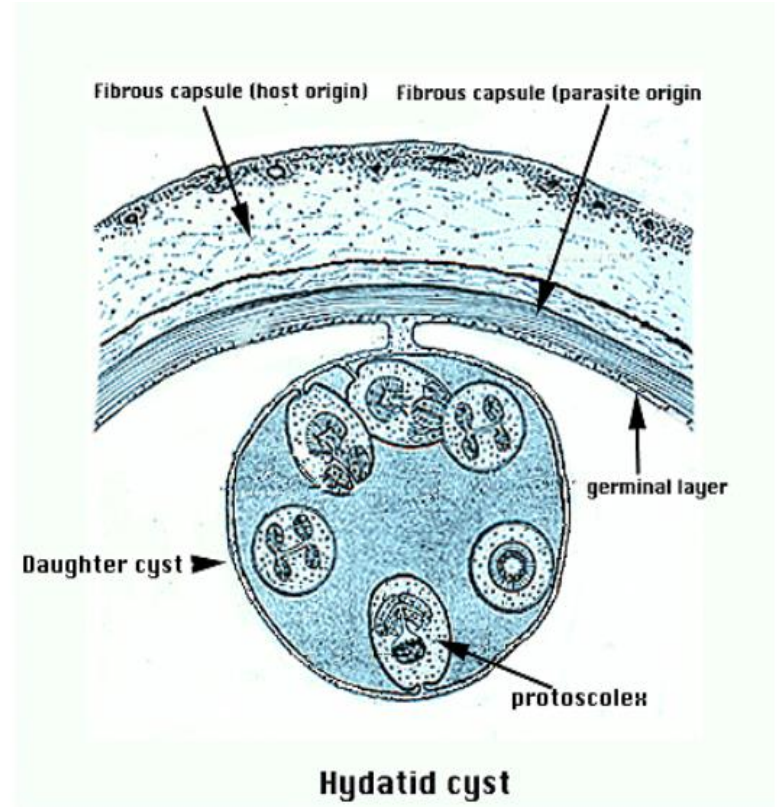
Under the microscope, the hydatid cysts are recognizable by three distinct characteristics

1. Thick, a cellular, laminated layer with characteristic acidophilic staining
2. Cellular germinal layer
3. Brood capsules or protoscolices

A host-produced granulomatous reaction can be seen surrounding the cyst

These cysts have an inner germinal layer and an outer laminated layer surrounded by a fibrous capsule derived from the host.

Smaller "daughter" cysts bud from the inner cellular layer.



Disease progress

The cysts grow rather slowly and are separated from the surrounding tissues by peculiar layers, so that the immune system of the host takes no or only little notice of the growing cyst.

1/ Non-fluid-filled: tubes are produced inside intermediate hosts, in the case of infections after ingestion of eggs of *E. multilocularis* and some related species

2/ At the inner wall of these hollow tubes, protoscolices are formed, which grow up to final adult worms as soon as the final hosts have ingested these stages

Symptoms of the disease (Echinococcosis, Echinococciasis):

Most patients with echinococcosis are asymptomatic, the infection being detected incidentally on an imaging study. Symptoms generally develop when the hydatid cyst reaches a size of 8 to 10 cm and begins when compressing vital structures or eroding into the biliary tract or a pulmonary bronchus

The cysts can also become superinfected, resulting in a bacterial abscess. Cyst leakage or rupture can result in an anaphylactic reaction, causing fever and hypotension.

Cysts can also develop in the brain, heart, kidneys, eyes, and bones.

The nature of signs and symptoms varies depending on the site of the cyst:

1/ liver : greater than 65%

2/ lungs : (25%)

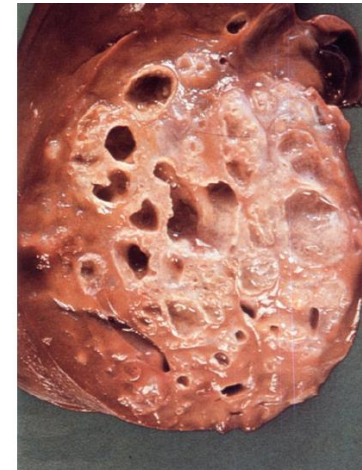
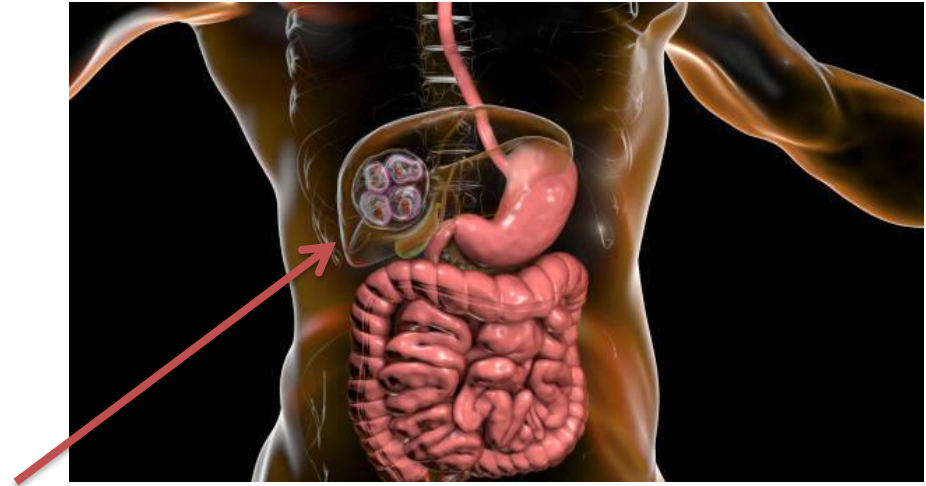
3/ other sites, including bones, spleen, central nervous system, and the heart.

Most primary infections consist of one cyst, but 20 to 40% of patients display multiple organ involvement.

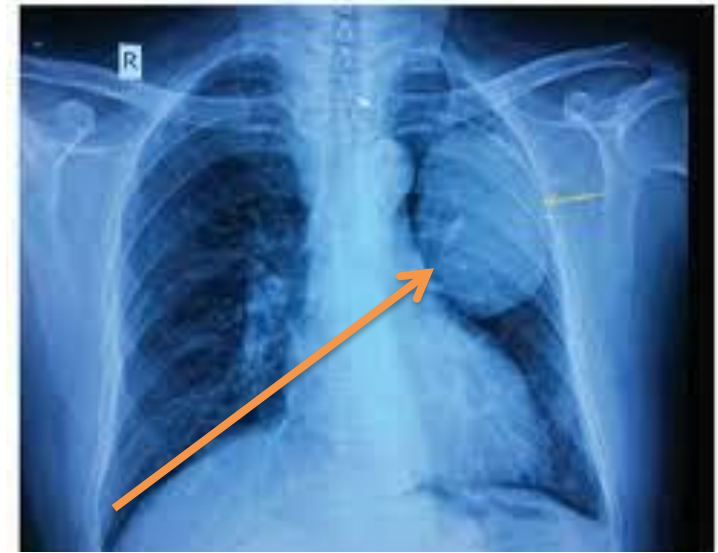
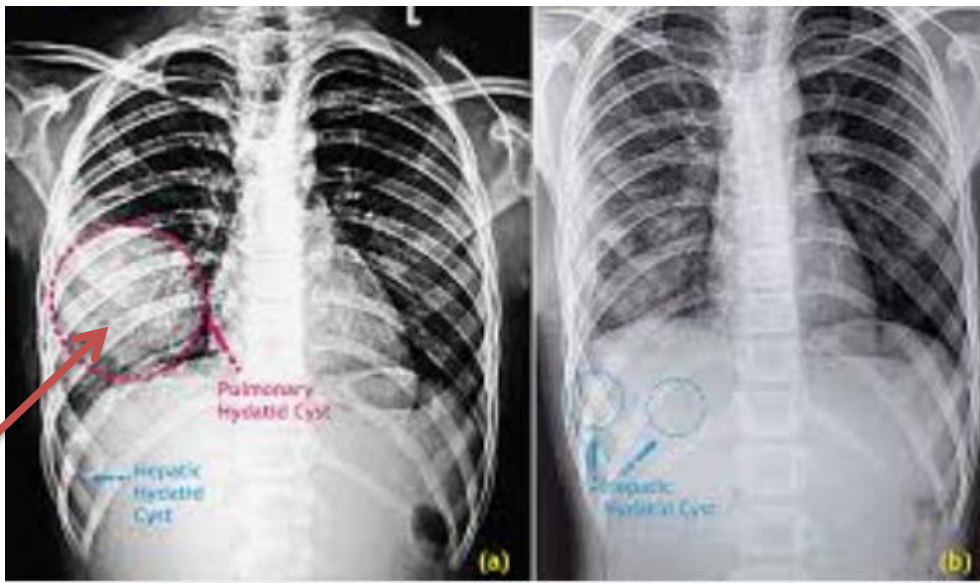
Infections can be acquired in childhood; most causes of hydatidosis present later in life due to the slow-growing nature of cysts.

However, cysts in the brain or eye can cause symptoms early on and thus mostly present in childhood.

If a cyst continues growing, patients may exhibit symptoms gradually due to the cyst exerting pressure on surrounding tissue. Sudden signs and symptoms are more likely due to rupture rather than the growth of the cyst. Common findings in liver hydatidosis include abdominal pain, decreased appetite, hepatomegaly, a palpable mass, and abdominal distention.



Human liver showing the cross-sectioned hollow tubes of a cyst system due to an infection with *Echinococcus multilocularis*



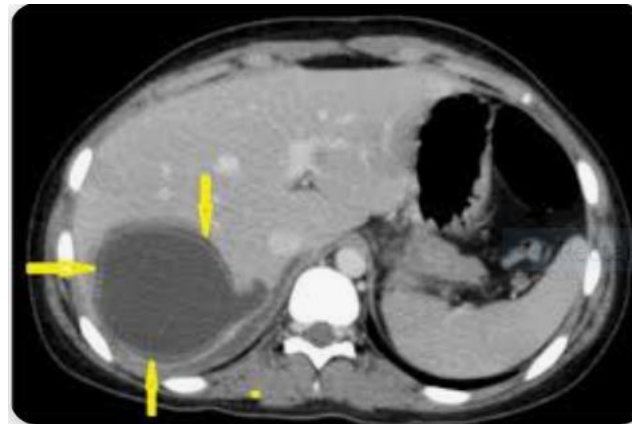
In the case of lung stages, pleural fluids may introduce collapse of the lungs , common findings in lung hydatidosis include chronic cough, chest pain, and shortness of breath.

while stages in the brain are characterized by severe symptoms like neuromotor deficiencies (e.g., paralyses). In the case that the cysts occur inside bones, fractures are common. The rare cases of ocular infestations lead to a so-called exophthalmos—protruding eyes.

Diagnosis :

The primary diagnosis in humans is based on the detection of cystic structures in different organs (liver, lung, brain, uterus, etc.). This can be done with the help of computed tomography, X-ray techniques, sonography, etc. These techniques do not allow a species-specific diagnosis.

However, species determination is very important, since the fine tubular structures of *E. multilocularis* contain omnipotent cells, which would be spread if such a structure is cut through during surgery.



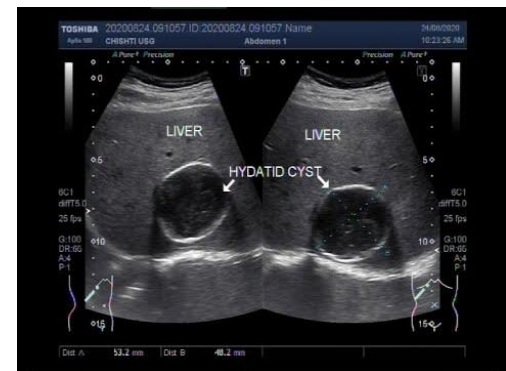
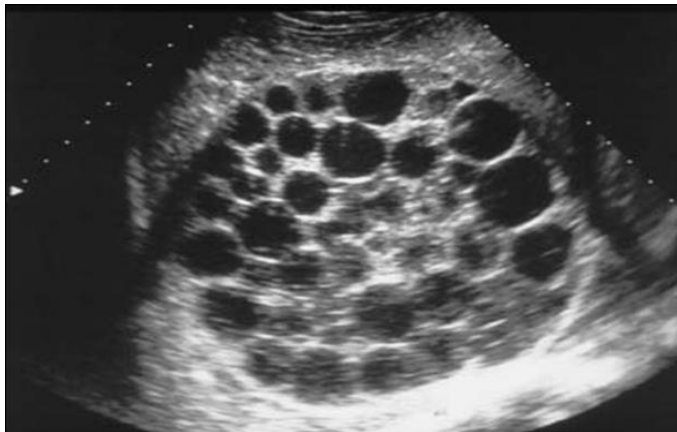
2/ The complete blood count may show eosinophilia

3/ Serological testing for Echinococcus antibodies is a common method for diagnosis. ELISA tests are usually used to screen

4/ confirmatory immunoblot assay for echinococcal antigens

5/ Liver function tests are unreliable in detecting the severity of disease and are abnormal in only about 40% of patients. Alkaline phosphatase is the one typically elevated, while AST, ALT, and bilirubin levels typically remain within normal range.

6/ Ultrasonography is the preferred modality for liver and intraabdominal hydatidosis. It remains the modality of choice for screening due to its accessibility and portability. It's also helpful in post-treatment monitoring of the disease.



Treatment

- In the case of the bladder-like large cysts (hydatids) due to infections with *E. granulosus*, surgical removal is possible depending on the position of the cyst inside the infected organ,
- whereas in the case of *E. multilocularis*, surgical removal is very difficult, since these stages end in very fine, practically invisible strands. In these cases, surgery would be a big mistake, since single undifferentiated cells would remain and—like in cancer—the cell divisions would go on.
- albendazole (12–15 mg/kg bodyweight) or mebendazole (50–60 mg/kg bodyweight) led to good survival rates. Also, in cases of the successful removal of a hydatid cyst, it is recommended to go on with the drug application for at least 2 years.

Medical treatment:

Benzimidazoles (BMZ) is indicated for:

- 1/ cysts smaller than 5cm in the liver and lung
- 2/ patients with cysts in two or more organs
- 3/ inoperable patients.

BMZs are also used following surgery or percutaneous procedures to prevent recurrence.

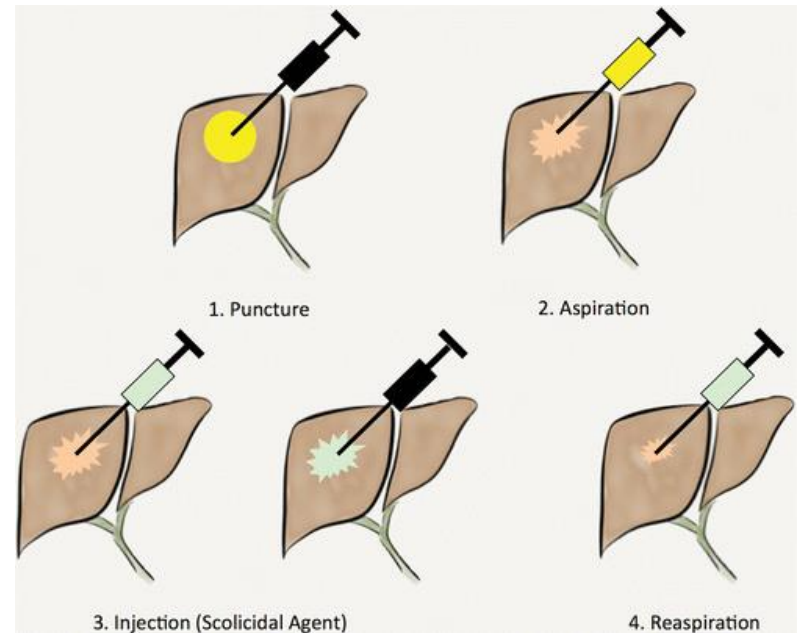
Albendazole (ABZ) is currently the drug of choice at a dose of 12 to 15 mg/kg/day.

Mebendazole (MBZ) is another therapeutic option. Medical treatment with BMZ is contraindicated in cysts vulnerable to rupture and in early pregnancy

Surgery is the modality of choice for:

- 1/ complicated cysts (large cysts)
- 2/ superficial cysts that may rupture with trauma or spontaneously
- 3/ infected cysts
- 4/ cysts with biliary tree communication
- 5/ cysts exerting pressure on adjacent organs.

PAIR (puncture, aspiration, injection, re-aspiration) procedure is an ultrasound-guided, minimally invasive modality used in hepatic and other abdominal cysts. Indications include inoperable patients, relapsing cases after surgery, and failure to respond to medical treatment



Complications

Rupture of the cyst, whether spontaneous or traumatic, can cause site-specific complications.

Rupture of liver cysts into the biliary tree can result in :

1/biliary obstruction

2/superinfection of the cyst

3/ secondary peritonitis

while rupture of the lung cysts into the bronchial tree can result in

1/ pneumonitis

2/ pneumothorax

3/ pleural effusion

4/ secondary pleuritis

Epidemiology

- Due to the nature of the pathogen, hydatidosis is more predominant in communities that keep dogs for guarding and herding cattle.
- Geographically, the disease has a worldwide distribution with higher prevalence in the Mediterranean, Russia, China, North and East Africa, Australia, and South America
- **Prophylaxis:** Avoidance of contact with living or dead foxes, cats, and dogs of unknown origin. Dogs and cats should be protected by regular application of antiworm compounds, which act well against cestodes and nematodes.
- Hunters should always wear boots, which allow hot cleaning, and use one-way gloves when it is needed to investigate the fur of wild animals or when skinning foxes to get their fur.